

Docker Assignment



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Assignment 1

- Install Docker and Docker Compose on your machine.
- Verify the installation by running a test container.
- Create a simple web application (e.g., a Python Flask, DotNet, Java or Node application).
- Write a Dockerfile to containerize the application.
- Build the Docker image and run a container from it.
- Use Docker commands to list, start, stop, and remove containers.
- Inspect running containers and view logs.
- Write a docker-compose.yml file to define a multi-container application.
- Use Docker Compose to bring up the application and ensure all services are running correctly.
- Create a private Docker registry or use Docker Hub.
- Push your Docker images to the registry.
- Pull the images from the registry and run them locally.

Requirements

- Docker 20.10 or later
- Docker Compose 1.29 or later
- Linux command-line knowledge
- GitHub account

Note

- Complete the tasks as specified.
- Submit your project in a GitHub repository.
- Ensure your submission is well-documented.
- Include a README file with instructions on how to set up and run your project.

Deliverables

- GitHub repository link with all source code and documentation.
- README file with setup instructions.
- Well-documented code with comments.
- Dockerfiles, DockerCompose files, and CICD configuration files.

Bonus Question

- Extend the multi-container application to include a database service and configure the web application to interact with the database.